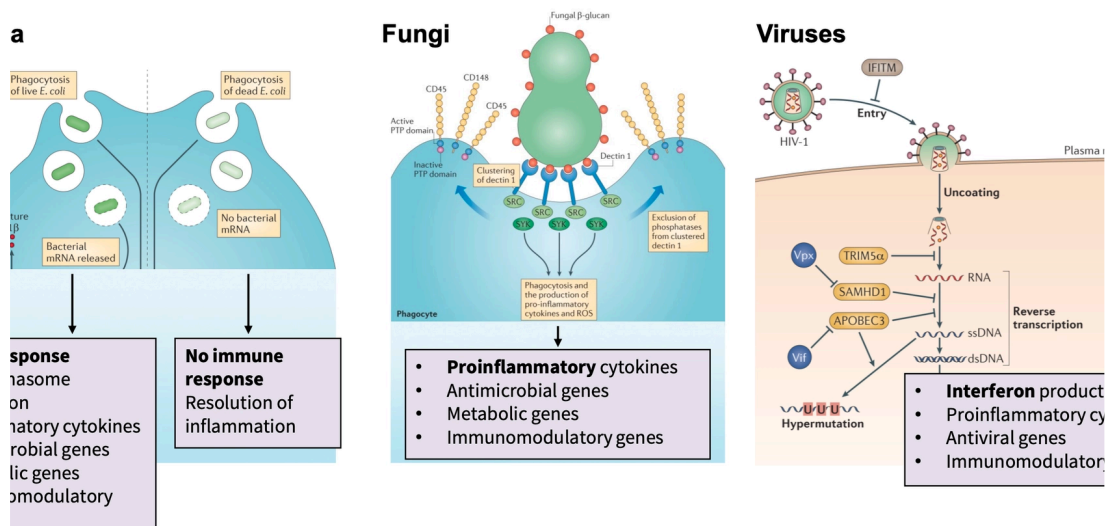


4.14 Immune Response to Infection

- ▼ Recall three microbe associated molecular patterns (MAMPs).
 - **Lipopolysaccharides** from **gram-negative bacteria**
 - **Peptidoglycan** from all **bacteria**
 - **Beta-glucan cell wall** on **fungi**
 - **Surface glycoproteins** and **nucleic acids** from **viruses**
- ▼ Recall three pattern/danger associated molecular patterns (PAMPs/DAMPs).
 - **ATP** released from **dead cells**
 - **Alarmins** - chemicals released as a result of cell death
 - **Specific cytokines/interleukins**
- ▼ Why are natural killer cells described to be part of the innate immune response, even though they are lymphocytes?
 - Because they **respond very quickly** and in an **antigen-independent manner**
- ▼ What is an advantage of the adaptive immune system over the innate?
 - A lot **more microbe-associated molecular patterns** (antigens) can be detected by the **adaptive system**
 - About **1000 molecular patterns** are **detected by the innate** in comparison to **$\sim 10^7$ by the adaptive**
- ▼ Is blood sterile (lacks viruses and bacterial presence)?
 - Yes. Blood is completely sterile and you would be in the ICU if it wasn't.
- ▼ In the immune response, which are the first types of cells to reach the site of infection?
 - **Neutrophils**
 - They are **followed by macrophages** after neutrophilic release of cytokines.

- ▼ What happens to neutrophils and macrophages when they interact with pathogens?
 - They become activated
 - This is because they are normally naïve as they come from bone marrow, but their gene expression changes after interaction with pathogens.
- ▼ How can the immune system distinguish between dead and live pathogens?
 - Dead pathogens have **much less RNA** than live pathogens
- ▼ Image showing the different molecules produced in response to different pathogens.

phagocyte responses are pathogen-specific

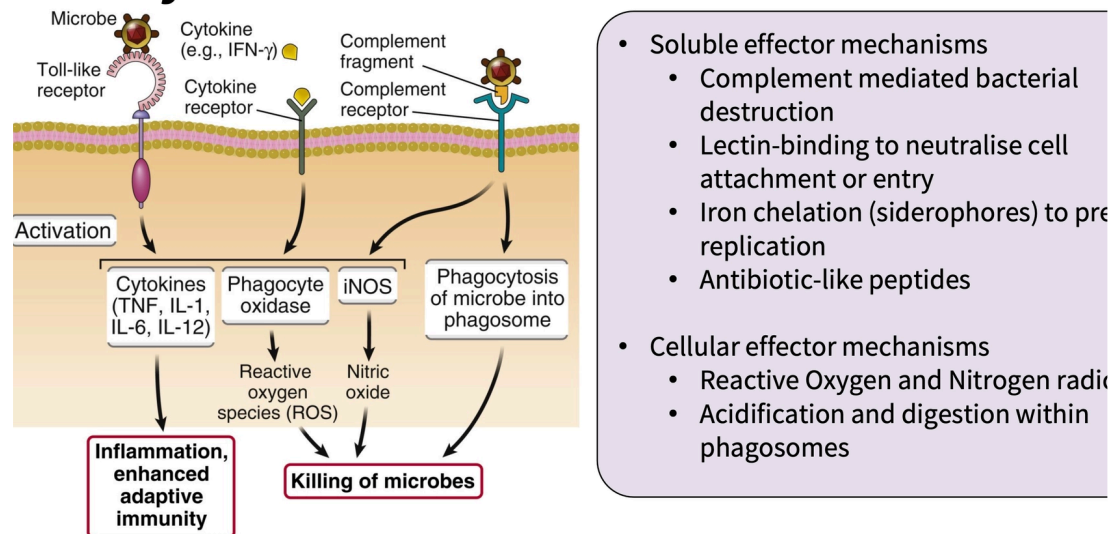


sequence and timing of the immune response to infection

- ▼ Describe how interferon- γ can lead to the activation of macrophages.
 - The **macrophage binds to a pathogen** and releases cytokines, which activates **lymphocytes**
 - **Lymphocytes** produce **interferon- γ**
 - **Interferon- γ** is important as it can **activate the macrophage**, allowing it to cause **enhanced** phagocytosis, increased cytokine production, expression of cell surface molecules, and antigen presentation (hence T cell activation)
 - It also allows the **macrophage to express certain antimicrobial genes** for molecules such as **reactive oxygen species** (i.e bleach for killing cells)
 - **Interferon- γ** is hence vital to controlling infection and the transition from the innate to adaptive immune response
- ▼ Which type of interferons are used only against viruses?
 - **Type I (IFN α/β)**
 - **Type II (IFN γ)** and **Type III (IFN λ)** also work against viruses but have other functions too (**Type II** = antibacterial, Th1 skewing | **Type III** = antiviral, mucosal immunity)
- ▼ Why do interferons have their name?
 - Because they were discovered to **interfere with viral replication**.
- ▼ What is pyroptosis?
 - A type of cell death which involves the influx of water into cells via their pores which causes them to burst
 - This leads to **inflammation** as well as **fevers (increased body temperature)**
- ▼ Describe how Natural Killer (NK) cells are involved in the immune response.
 - Natural Killer cells **detect infected cells** and this leads to them triggering **apoptosis in the infected target cell**.

- ▼ Image showing humoral and cellular innate immune responses.

Combining humoral & cellular innate immunity



- As seen here, the complement system can be used to **opsonise bacteria (using C3b)** which leads to **enhanced phagocytosis**, or it can produce **reactive oxygen species**, which lead to their **lysis (through MAC)**.

- ▼ Which two cytokines work antagonistically in bacterial and fungal infections?

- **Interferon-γ (IFNγ)** is used for **bacterial infections**
- **IL-17** is used for **fungal infections**

- ▼ Recall which cytokines are involved in Th1, Th2 and Th17 immunity.

- **Th1** - IFN-γ
- **Th2** - IL-4, IL-5, IL-13
- **Th17** - IL-17, IL-22

- ▼ Recall which type of leukocyte are targeted by these T helper cells and which pathogens each of these T cells are produced against.

- **Th1** - **Macrophage** - intracellular pathogens (i.e. viruses)
- **Th2** - **Eosinophils** - Helminths, parasites
- **Th17** - **Neutrophils** - extracellular bacteria and fungi

- ▼ Table showing how different phagocytes release different cytokines which lead to different lymphocytes being produced.

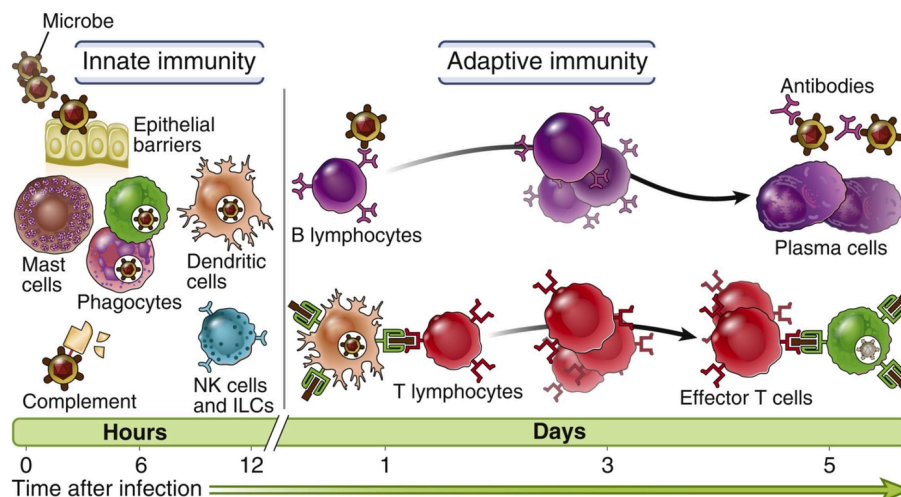
Microbe-specific phagocyte responses induce the appropriate lymphocyte phase



Effector T cells	Defining cytokines	Principal target cells	Major immune reactions	Host defense	Role in disease
Th1	IFN- γ	Macrophages	Macrophage activation	Intracellular pathogens	Autoimmunity; chronic inflammation
Th2	IL-4 IL-5 IL-13	Eosinophils	Eosinophil and mast cell activation; alternative macrophage activation	Helminths	Allergy
Th17	IL-17 IL-22	Neutrophils	Neutrophil recruitment and activation	Extracellular bacteria and fungi	Autoimmunity; inflammation

- ▼ Image showing the sequence and timing of events in the cellular immune response.

Timing of the cellular immune response



- As we can see here, the immune response progresses from complement and phagocytes within a few hours, to B and T lymphocytes after about two days, to effector T cells and plasma cells (and hence antibodies) after about five days

